Stored Procedures Assignment solutions

1. delimiter //
   mysql> create procedure check_even_odd (in num INT)
   -> begin
   -> if num%2 = 0 then
   -> select 'Even' as result;
   -> else
   -> select 'Odd' as result;
   -> end if;
   -> end //
   DELIMITER ;
   mysql> CALL check_even_odd(25);
   +--------+
   | result |
   +--------+
   | odd |
   +--------+
   1 row in set (0.00 sec)
   Query OK, 0 rows affected (0.01 sec)
   mysql> CALL check_even_odd(40);
   +--------+
   | result |
   +--------+
   | even |
   +--------+
   1 row in set (0.00 sec)
   Query OK, 0 rows affected (0.00 sec)

2. delimiter //
mysql> create procedure calculate_interest(
   -> IN principal INT,
-> IN rate INT,
   -> IN time INT
   ->)
   -> begin
-> declare v_interest decimal(10,2);

```
    -> set v_interest = (principal * rate * time)/100;
    -> select v_interest as Simple_Interest;
    -> END //
    DELIMITER ;
    mysql> call calculate_interest (10000, 5, 2);
    +-----------------+
    | Simple_Interest |
+-----------------+
|       1000.00 |
    +-----------------+
    1 row in set (0.01 sec)
Query OK, 0 rows affected (0.01 sec)
call calculate_interest (25000, 7, 3);
+-----------------+
| Simple_Interest |
+-----------------+
|       5250.00 |
+-----------------+
1 row in set (0.01 sec)
Query OK, 0 rows affected (0.01 sec)
```

3. DELIMITER //
   mysql> create procedure calculate_grade (IN studentID INT,
   student_name VARCHAR(40), marks INT)
   -> begin
   -> if marks>=90 and marks<=100 then
   -> select 'A+' as grade;
   -> elseif marks>=80 and marks<=89 then
   -> select 'A' as grade;
   -> elseif marks>=70 and marks<=79 then
   -> select 'B' as grade;
   -> elseif marks>=60 and marks<=69 then
   -> select 'C' as grade;
   -> elseif marks>=50 and marks<=59 then
   -> select 'D' as grade;
   -> else
   -> select 'F' as grade;
   -> end if;
   -> end //
   CALL calculate_grade(101, 'Amit', 85);
   +-------+

```
| grade |
+-------+
| A     |
+-------+
1 row in set (0.01 sec)
Query OK, 0 rows affected (0.01 sec)
mysql> CALL calculate_grade(102, 'Priya', 72);
+-------+
| grade |
+-------+
| B     |
+-------+
1 row in set (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
mysql> CALL calculate_grade(103, 'Rahul', 45);
+-------+
| grade |
+-------+
| F     |
+-------+
1 row in set (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
```

4. DELIMITER //

```
mysql>
mysql> CREATE PROCEDURE calculate_bonus(
  ->     IN employee_name VARCHAR(40),
  ->     IN salary DECIMAL(10,2)
  -> )
  -> BEGIN
  ->     DECLARE bonus_percent INT;
  ->     DECLARE bonus_amount DECIMAL(10,2);
  ->     DECLARE total_salary DECIMAL(10,2);
  ->
  ->     IF salary >= 100000 THEN
  ->         SET bonus_percent = 20;
  ->     ELSEIF salary >= 50000 THEN
  ->         SET bonus_percent = 15;
  ->     ELSEIF salary >= 30000 THEN
  ->         SET bonus_percent = 10;
  ->     ELSE
```

```
  ->       SET bonus_percent = 5;
  ->    END IF;
  ->
  ->    SET bonus_amount = salary * bonus_percent / 100;
  ->    SET total_salary = salary + bonus_amount;
  ->
  ->    SELECT employee_name, salary, bonus_percent,
  ->          bonus_amount, total_salary;
  -> END //
Query OK, 0 rows affected (0.02 sec)
mysql> DELIMITER ;
mysql> call calculate_bonus('Amit', 120000);
+---------------+-----------+---------------+--------------+--------------+
| employee_name | salary    | bonus_percent | bonus_amount |
total_salary |
+---------------+-----------+---------------+--------------+--------------+
| Amit          | 120000.00 |          20 |    24000.00 |    144000.00 |
+---------------+-----------+---------------+--------------+--------------+
1 row in set (0.01 sec)
Query OK, 0 rows affected (0.02 sec)

mysql> call calculate_bonus('Priya', 65000);
+---------------+----------+---------------+--------------+--------------+
| employee_name | salary   | bonus_percent | bonus_amount | total_salary
|
+---------------+----------+---------------+--------------+--------------+
| Priya         | 65000.00 |          15 |     9750.00 |     74750.00 |
+---------------+----------+---------------+--------------+--------------+
1 row in set (0.00 sec)
Query OK, 0 rows affected (0.01 sec)

mysql> call calculate_bonus('Rahul', 25000);
+---------------+----------+---------------+--------------+--------------+
| employee_name | salary   | bonus_percent | bonus_amount | total_salary
|
+---------------+----------+---------------+--------------+--------------+
| Rahul         | 25000.00 |          5 |     1250.00 |     26250.00 |
+---------------+----------+---------------+--------------+--------------+
1 row in set (0.00 sec)
Query OK, 0 rows affected (0.01 sec)
```

5. DELIMITER //

```
mysql>
mysql> CREATE PROCEDURE find_maximum(
    ->     IN num1 INT,
    ->     IN num2 INT,
    ->     IN num3 INT
    -> )
    -> BEGIN
    ->     DECLARE largest_num INT;
    ->
    ->     IF num1 >= num2 AND num1 >= num3 THEN
    ->         SET largest_num = num1;
    ->     ELSEIF num2 >= num1 AND num2 >= num3 THEN
    ->         SET largest_num = num2;
    ->     ELSE
    ->         SET largest_num = num3;
    ->     END IF;
    ->
    ->     SELECT largest_num AS maximum_number;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL find_maximum(10, 25, 15);
+----------------+
| maximum_number |
+----------------+
|             25 |
+----------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL find_maximum(25, 78, 45);
+----------------+
| maximum_number |
+----------------+
|             78 |
+----------------+
1 row in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)

mysql> CALL find_maximum(100, 50, 75);
+----------------+
| maximum_number |
+----------------+
|          100 |
+----------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL find_maximum(30, 30, 20);
+----------------+
| maximum_number |
+----------------+
|           30 |
+----------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

6. DELIMITER //

```
mysql>
mysql> CREATE PROCEDURE salary_analysis(
    ->     IN employee_name VARCHAR(40),
    ->     IN salary DECIMAL(10,2)
    -> )
    -> BEGIN
    ->     DECLARE category VARCHAR(20);
    ->     DECLARE allowance_percent INT;
    ->     DECLARE allowance_amount DECIMAL(10,2);
    ->     DECLARE final_salary DECIMAL(10,2);
    ->
    ->     IF salary >= 100000 THEN
    ->         SET category = 'High';
    ->         SET allowance_percent = 20;
    ->     ELSEIF salary >= 50000 THEN
    ->         SET category = 'Medium';
    ->         SET allowance_percent = 15;
```

```
  ->     ELSE
  ->         SET category = 'Low';
  ->         SET allowance_percent = 10;
  ->     END IF;
  ->
  ->     SET allowance_amount = salary * allowance_percent / 100;
  ->     SET final_salary = salary + allowance_amount;
  ->
  ->     SELECT employee_name, salary, category,
  ->            allowance_percent, allowance_amount, final_salary;
  -> END //
Query OK, 0 rows affected (0.02 sec)

DELIMITER ;
mysql> CALL salary_analysis('Amit', 120000);
+---------------+-----------+----------+-------------------+-----------------+-------------+
| employee_name | salary    | category | allowance_percent | allowance_amount | final_salary |
+---------------+-----------+----------+-------------------+-----------------+-------------+
| Amit          | 120000.00 | High     |            20 |       24000.00 |    144000.00 |
+---------------+-----------+----------+-------------------+-----------------+-------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL salary_analysis('Priya', 60000);
+---------------+----------+----------+-------------------+-----------------+-------------+
| employee_name | salary   | category | allowance_percent | allowance_amount | final_salary |
+---------------+----------+----------+-------------------+-----------------+-------------+
| Priya         | 60000.00 | Medium   |            15 |        9000.00 |     69000.00 |
+---------------+----------+----------+-------------------+-----------------+-------------+
1 row in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.01 sec)

mysql> CALL salary_analysis('Rahul', 35000);
+---------------+----------+----------+------------------+-----------------+-------
--------+
| employee_name | salary   | category | allowance_percent |
allowance_amount | final_salary |
+---------------+----------+----------+------------------+-----------------+-------
--------+
| Rahul         | 35000.00 | Low      |             10 |        3500.00 |    38500.00 |
+---------------+----------+----------+------------------+-----------------+-------
--------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

7. DELIMITER //

```
mysql>
mysql> CREATE PROCEDURE calculate_electricity_bill(
  ->    IN consumer_name VARCHAR(40),
  ->    IN units INT
  -> )
  -> BEGIN
  ->    DECLARE total_bill DECIMAL(10,2);
  ->
  ->    IF units <= 100 THEN
  ->       SET total_bill = units * 2;
  ->
  ->    ELSEIF units <= 200 THEN
  ->       SET total_bill = (100 * 2) + ((units - 100) * 3);
  ->
  ->    ELSEIF units <= 400 THEN
  ->       SET total_bill = (100 * 2) + (100 * 3)
  ->                   + ((units - 200) * 5);
  ->
  ->    ELSE
  ->       SET total_bill = (100 * 2) + (100 * 3)
  ->                   + (200 * 5) + ((units - 400) * 7);
  ->    END IF;
  ->
```

```
  ->    SELECT consumer_name, units AS units_consumed,
  ->           total_bill;
  -> END //
Query OK, 0 rows affected (0.02 sec)
DELIMITER ;
mysql>
mysql> CALL calculate_electricity_bill('Amit', 80);
+---------------+----------------+------------+
| consumer_name | units_consumed | total_bill |
+---------------+----------------+------------+
| Amit          |             80 |     160.00 |
+---------------+----------------+------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL calculate_electricity_bill('Priya', 150);
+---------------+----------------+------------+
| consumer_name | units_consumed | total_bill |
+---------------+----------------+------------+
| Priya         |            150 |     350.00 |
+---------------+----------------+------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL calculate_electricity_bill('Rahul', 350);
+---------------+----------------+------------+
| consumer_name | units_consumed | total_bill |
+---------------+----------------+------------+
| Rahul         |            350 |    1250.00 |
+---------------+----------------+------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL calculate_electricity_bill('Sneha', 500);
+---------------+----------------+------------+
| consumer_name | units_consumed | total_bill |
+---------------+----------------+------------+
| Sneha         |            500 |    2200.00 |
```

```
+---------------+----------------+------------+
```

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)